

QuestionsUnit3-LeafVSTree.pdf



abcd__1234



Paralelismo



3º Grado en Ingeniería Informática



Facultad de Informática de Barcelona (Fib)
Universidad Politécnica de Catalunya



[Accede al documento original](#)

 **Gemini 3**
Nuestro modelo más preciso

Sintetiza horas de
Investigación en minutos

Necesito estudiar a fondo el comportamiento de la fotosíntesis según el tipo de planta y el entorno



 Deep Research 



 Fuentes 

 Archivos

Estudia con Gemini

Gemini 3 Nuestro modelo más preciso

Necesito preparar un test para autoevaluarme sobre este archivo

Apuntes Tema 3

PDF

+ Herramientas

Razonamiento

Estudia con Gemini

Qualificació 9,00 sobre 9,00 (100%)

Pregunta 1

Correcte

Puntuació 1,00
sobre 1,00

Marca la
pregunta

For the following parallel computation expressed using tasks

```
#define N 1024
#define MIN 16

void doComputation (int * vector, int n) {
    int size = n / 4;
    for (int i = 0; i < n; i += 4)
        tareador_start_task("compute");
        compute(&vector[i], size);
        tareador_end_task("compute");
}

void partition (int * vector, int n) {
    if (n > MIN) { // MIN is multiple of 4
        int size = n / 4;
        for(int i=0; i<4; i++)
            partition(&vector[i*size], size);
        }
    else
        doComputation(vector, n);
    return;
}

void main() {
    ...
    partition (vector, N); // N is multiple of 4
    ...
}
```

How many tasks are generated during the execution of the program?

Resposta:

256



256 tasks will be generated, where each task will work on 4 elements of the vector.

Pregunta 2

Correcte

Puntuació 1,00
sobre 1,00 [Marca la pregunta](#)

If the granularity of a task is defined as the number of invocations to function `compute` (e.g. granularity 4 means that tasks perform 4 invocations of that function). Which is the granularity for the tasks in this task decomposition?

Resposta: 

Each task performs 1 invocation to function `compute`.

Pregunta 3

Correcte

Puntuació 1,00
sobre 1,00 [Marca la pregunta](#)

Assuming that the execution of each `compute` invocation takes 16 time units, the creation of each individual task takes 2 time units and that we neglect the execution time of the rest of the code, calculate the parallel execution time for an infinite number of processors in the system (T_{∞}).

Resposta: 

Pregunta 4

Correcte

Puntuació 1,00
sobre 1,00

 Marca la
pregunta

If the task decomposition is changed to

```
#define N 1024
#define MIN 16

void doComputation (int * vector, int n) {
    int size = n / 4;
    for (int i = 0; i < n; i += 4)
        compute(&vector[i], size);
}

void partition (int * vector, int n) {
    if (n > MIN) { // MIN is multiple of 4
        int size = n / 4;
        for(int i=0; i<4; i++)
            partition(&vector[i*size], size);
    }
    else
        tareador_start_task("doComputation");
        doComputation(vector, n);
        tareador_end_task("doComputation");
    return;
}

void main() {
    ...
    partition (vector, N); // N is multiple of 4
    ...
}
```

How many explicit tasks are generated during the execution of the program?

Resposta: 64



WUOLAH

2020
26
8-11 JULY

MAD
COOL
FESTIVAL

CONSIGUE
TU ENTRADA
AQUÍ →



Pregunta 5

Correcte

Puntuació 1,00
sobre 1,00

[Marca la pregunta](#)

If the granularity of a task is defined as the number of invocations to function `compute` (e.g. granularity 4 means that tasks perform 4 invocations of that function). Which is the granularity for the tasks in this task decomposition?

Resposta: ✓

Pregunta 6

Correcte

Puntuació 1,00
sobre 1,00

[Marca la pregunta](#)

Assuming that the execution of each `compute` invocation takes 16 time units, the creation of each individual task takes 2 time units and that we neglect the execution time of the rest of the code, calculate the parallel execution time for an infinite number of processors in the system (T_{∞}).

Resposta: ✓

Well done!

WUOLAH

Pregunta 7

Correcte

Puntuació 1,00
sobre 1,00

Marca la
pregunta

Finally, if the task decomposition is changed to

```
#define N 1024
#define MIN 16

void doComputation (int * vector, int n) {
    int size = n / 4;
    for (int i = 0; i < n; i += 4)
        compute(&vector[i], size);
}

void partition (int * vector, int n) {
    if (n > MIN) { // MIN is multiple of 4
        int size = n / 4;
        for(int i=0; i<4; i++)
            tareador_start_task("partition");
            partition(&vector[i*size], size);
            tareador_end_task("partition");
        }
    else
        doComputation(vector, n);
    return;
}

void main() {
    ...
    partition (vector, N); // N is multiple of 4
    ...
}
```

How many explicit tasks are generated during the execution of the program?

Resposta: 84





8-11 JULY 2026

8th WEDNESDAY

FOO FIGHTERS · MOBY · THE WAR ON DRUGS · WOLF ALICE
THE LAST DINNER PARTY · THE WARNING · JEHNNY BETH · THE WAR AND TREATY
PALAYE ROYALE · DOGSTAR · HOT MILK · VILLANELLE · HOTWAX
HOONINE · MADMADMAD · BIGGER SPLASH

9th THURSDAY

FLORENCE + THE MACHINE · JENNIE · LORDE · TEDDY SWIMS
ZARA LARSSON · RENÉE RAPP · CHARLIE PUTH · THE BLAZE dj set · CMAT
SADIE JEAN · ZIMMER90 · CHLOE SLATER · FROST CHILDREN · SON MIEUX · FLORENTENES · LA PALOMA
BOYS NOIZE · PALMS TRAX · YUNG PRADO · AMRITA THE LOOP

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POLO & PAN dj set · BUNT. · SWIMMING PAUL · WEVAL · MAESIC THE LOOP

11th SATURDAY

NICK CAVE & THE BAD SEEDS · PULP · THE BLACK CROWES · DAVID BYRNE
KASABIAN · JALEN NGONDA · THE VACCINES · MATT BERNINGER
THE REYTONS · PERSIA HOLDER · OVERPASS · TTSSFU · SUPERMODEL*
RICHIE HAWTIN · NINA KRAVIZ · AEREA · LUXI VILLAR THE LOOP

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Pregunta 8

Correcte

Puntuació 1,00
sobre 1,00 [Marca la pregunta](#)

If the granularity of a task is defined as the number of invocations to function `compute` (e.g. granularity 4 means that tasks perform 4 invocations of that function). Which is the granularity for those tasks in this task decomposition that end up executing instances of function `compute`?

Resposta: **Pregunta 9**

Correcte

Puntuació 1,00
sobre 1,00 [Marca la pregunta](#)

Assuming that the execution of each `compute` invocation takes 16 time units, the creation of each individual task takes 2 time units and that we neglect the execution time of the rest of the code, calculate the parallel execution time for an infinite number of processors in the system (T_{∞}).

Resposta: 

Well done!

WUOLAH